

epidesc: An R package for computing epidemiological descriptors to characterise infectious disease transmission patterns

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Summary

The R package `epidesc` provides a standardised pipeline for computing epidemiological descriptors (epi-descriptors) from weekly case count time series of infectious diseases. These epi-descriptors characterise the transmission dynamics, by quantifying the magnitude, duration, and temporal patterns of cases at a spatial unit, enabling the classification of areas into distinct transmission patterns (e.g., persistent, epidemic, episodic, outbreaks, trends). The workflow requires three steps: formatting temporal data into epidemiological year-week identifiers, selecting descriptors and their parameters from a built-in catalogue, and computing descriptors for each spatial unit and epidemiological year. `epidesc` is designed for researchers and public health professionals working on infectious disease surveillance, early warning systems, and spatial epidemiology.

Statement of need

Climate-sensitive diseases such as dengue, malaria, and leptospirosis exhibit heterogeneous transmission dynamics across space and time, driven by climatic, demographic, and socioeconomic factors [deAlmeida2022; Alcayna2025]. Studies have demonstrated that the spatio-temporal diffusion of dengue is conditioned by population mobility, vector distribution, and local social and environmental characteristics, producing distinct epidemic patterns even at the neighbourhood scale [Xavier2017]. Understanding and classifying these dynamics is essential for designing targeted surveillance strategies and allocating resources efficiently. Inspired by the epi-features calculated for influenza curves conceived by (Tabataba et al., 2017), deAlmeida2022 proposed a set of 13 epi-descriptors to characterise dengue transmission patterns in Brazilian municipalities, demonstrating their utility for clustering areas into persistent, epidemic, and episodic transmission categories. However, the original analysis relied on custom scripts that were not packaged for reuse by the broader research community.

`epidesc` addresses this gap by encapsulating the epi-descriptors computation methodology into a modular and flexible R package. The target audience includes epidemiologists, data scientists, and public health analysts who need to systematically characterise disease transmission patterns from routine notification data. The package facilitates the reproduction and extension of epi-descriptors, allows comparative patterns across diseases and geographies, and supports the development of public health interventions tailored to local transmission patterns.

State of the field

Several R packages support epidemiological time series analysis. Early tools like surveillance [Meyer2017] and EpiEstim [Cori2013] established standard frameworks for outbreak detection and real-time reproduction number (R_t) estimation. Accelerated by the COVID-19 pandemic, the wider Epiverse-TRACE software initiative has significantly expanded this toolkit. Modern packages include EpiNow2 [Abbott2026] for Bayesian estimation of infection trajectories and R_t under reporting delays, incidence2 [Taylor2024] for computing and visualising incidence curves, and specialized tools such as epiparameter [Lambert2025] for epidemiological parameter extraction. Outside the R ecosystem, EpiViewer [Thorve2018] provides a web-based framework for exploring, comparing, and organising epidemiological time series, with built-in computation of general epidemic features such as peak timing and magnitude, cumulative counts, and first take-off time. Epipoi is an useful epidemiological time series analysis tool for detecting trends, seasonality, anomalies and spatial patterns [Alonso2012]. It was developed and compiled in MATLAB but its last version is outdated. However, none of these packages provide a dedicated framework for computing the specific set of temporal descriptors needed to characterise and classify disease transmission patterns at a spatial unit.

The novelty of epidesc lies in its focus on descriptive features of epidemic time series rather than on mechanistic modelling or outbreak detection. By computing indicators such as the maximum duration of consecutive weeks with at least x cases, the proportion of weeks exceeding this threshold, and the frequency of case-free periods, epidesc enables a data-driven classification of transmission patterns. This approach complements existing tools and is particularly relevant for diseases with strong seasonality and spatial heterogeneity, such as dengue in tropical settings. Furthermore, epidesc is designed to accommodate the addition of new descriptors from the research community.

Software design

epidesc is structured as a standard R package following CRAN conventions, with comprehensive documentation via roxygen2 [Wickham2026], unit tests using testthat [Wickham2011], and a vignette demonstrating the complete workflow. The package depends on lubridate [Grolemund2011] for date manipulation and nseq [Saldanha2024] for efficient run-length encoding operations on time series.

epidesc is available on CRAN [10.32614/CRAN.package.epidesc](https://cran.r-project.org/web/packages/epidesc/index.html) and the development version can be accessed at <https://github.com/BSC-ES/epidesc>. The software architecture follows a three-step pipeline (Figure 1):



Figure 1: Schematic pipeline of the epidesc package. Input data containing weekly case counts with spatial and temporal identifiers are processed across three stages: date formatting with epiyearweek(), descriptor selection via desc_list(), and computation with desc_year(), producing epidemiological descriptors per spatial unit and epidemiological year.

Input data: As input, epidesc requires a dataframe containing epidemiological line-list data. This dataframe must include at least 3 columns: weekly case counts, a spatial identifier and a date.

Step 1 — Date formatting: the function epiyearweek() converts Date vectors into epidemiological year-week identifiers (format yyyyww), supporting both ISO weeks (starting Monday) and epidemiological weeks (starting Sunday).

80 *Step 2 — Descriptor specification:* users select descriptors from a built-in catalogue accessible
 81 via `desc_list()`, specifying parameters such as the minimum number of consecutive weeks or
 82 the case threshold. The catalogue currently includes 11 epi-descriptors organised into four
 83 classes (Table 1).

84 *Step 3 — Computation:* the main function `desc_year()` computes all specified epi-descriptors
 85 by spatial unit and epidemiological year.

86 *Output:* `epidesc` provides an object that stores selected epi-descriptors calculated by year or
 87 epidemiological year and spatial unit identifier.

88 Table 1- Epi-descriptors included in `epidesc` v0.1.0.

Class	Function	Description	Parameters
Peak	Ap	Maximum number of cases (peak) - amplitude	
Peak	Tp	Week where the maximum peak occurred - time	
Period with cases	Cnf	Frequency of periods of consecutive n weeks or longer with at least x cases	n, x
Period with cases	Cmax	Maximum duration in consecutive weeks with at least x cases	x
Period with cases	Cmed	Median duration in consecutive weeks with at least x cases	x
Period with cases	Isof	Number of weeks with isolated cases	
Period with cases	p	Proportion of weeks with at least x cases	x
Period without cases	Cwf	Frequency of periods of at least n consecutive weeks without cases	n
Period without cases	Cwmax	Maximum duration in consecutive weeks without cases	
Period without cases	Cwmed	Median duration in consecutive weeks without cases	
Incidence	Inc	Annual incidence per p population	p

89 With the exception of incidence rate calculations, the epi-descriptors do not require population
 90 estimates for the spatial unit of analysis. This facilitates their application in settings where
 91 population data are incomplete or not regularly updated, a common limitation at fine spatial
 92 scales and particularly relevant in resource-constrained settings.

93 The `desc_year()` function performs input validation by checking the regularity of the time series,
 94 identifying duplicate time points within spatial units, and verifying that population data are
 95 provided when incidence descriptor is requested. If the spatio-temporal grid is incomplete, the
 96 function issues a warning. Epi-descriptors are calculated only for complete epidemiological years;

if an epidemiological year has missing weeks or case records, the corresponding descriptors are returned as NA. Week 53, when present, is optionally handled by equally distributing its cases between weeks 52 of the same year and 01 of the next year, ensuring consistent 52-week years for descriptor calculation. This behaviour is controlled by the argument `collapse53`, which is TRUE by default. A minimal usage example is as follows:

```
library(epidesc)
data("dengueRio")

# step 1: date formatting
dengue <- dengueRio
dengue$yearweek <- epiyearweek(dengue$date, start = "Sunday")

# step 2: descriptor specification
descriptors <- list(
  Ap = list(fun = "Ap"),
  Cnf3 = list(fun = "Cnf", n = 3, x = 5),
  Inc = list(fun = "Inc", p = 100000)
)

# step 3: computation of epi-descriptors
res <- desc_year(
  data = dengue,
  cases = "cases",
  time = "yearweek",
  space = "muni_code",
  pop = "pop",
  sweek = 41,
  descriptors = descriptors
)
```

The package includes a sample dataset (`dengueRio`) containing weekly dengue cases for municipalities in the state of Rio de Janeiro (2017–2022), enabling users to explore the functionality immediately after installation.

Research impact statement

The methodological framework implemented in `epidesc` was originally developed and applied in @deAlmeida2022, which classified 1,823 Brazilian municipalities into four distinct dengue transmission patterns (persistent, epidemic, episodic/epidemic, and episodic) using data from 2010 to 2019. That study demonstrated the association between transmission patterns, population size, and climate, providing evidence for the existence of a critical community size for persistent dengue transmission. The classification framework has implications for the design of differentiated surveillance protocols and the achievement of World Health Organization goals for neglected tropical disease elimination.

By packaging this methodology into `epidesc`, we enable its application to other diseases (e.g., Zika, chikungunya, malaria), other geographies, and longer time periods. The package is currently being used within the Global Health Resilience (GHR) group at the Barcelona Supercomputing Center in Spain and at the Oswaldo Cruz Foundation (Fiocruz) in Brazil for ongoing research on climate-sensitive disease dynamics and early warning systems. The open-source availability of `epidesc` facilitates reproducibility and encourages community contributions of additional descriptors.

121 **Future developments and call for contributions**

122 In future versions of the package, we intend to expand the range of epi-descriptors to
123 include methodologies to characterise the seasonality of the disease. Furthermore, we call
124 for contributions to extend the list of supported epi-descriptors and welcome any suggestions
125 either via Github issues <https://github.com/BSC-ES/epidesc/issues> or by sending an email to
126 the corresponding authors.

127 **AI usage disclosure**

128 AI tools were used to code-review the package (in addition to human code review), to assist
129 with unit testing and support in English review.

130 **Author contributions**

131 RML, IFA, and CTC conceived and implemented epi-descriptors and curated the data demo.
132 CMG developed the R package. RML, CMG, and DSL contributed to package design. RML,
133 CMG, DSL, and DRX conducted tests and package validation. RML, CMG, IFA, DSL, DRX
134 and RL reviewed package GitHub repository. RML and DRX wrote the manuscript. All authors
135 reviewed the final manuscript version.

136 **Conflict of interest**

137 We declare we have no competing interests.

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